Currency Conversion Approach: Converting Budget and Actuals into INR

While reviewing the dashboard, I identified that the Actual and Budget values were available in multiple currencies such as INR, GBP, and USD. Since adding different currencies directly would give an incorrect consolidated number, I converted both Actuals and Budget into one reporting currency: INR.

1. FX Rate Table Preparation

I used the `FX_Rates` table as the source for currency conversion. Since the FX table had multiple daily rates for the same currency, I first standardized it before linking it to the financial data.

The key preparation steps were:

- Cleaned the `Currency` column by trimming spaces and keeping a consistent currency code format.
- Converted the date field into a proper date format.
- Created a `MonthStart` column from the FX date so that daily rates could be aligned with monthly Actual and Budget data.
- Grouped the FX table by `MonthStart` and `Currency`.
- Took the average FX rate for each currency-month combination.
- Created a clean monthly FX table named `FX_Rates_Monthly_Avg`.

The grouping logic was:

```
MonthStart + Currency
```

This helped create one average FX rate for each currency in each month.

For example:

MonthStart	Currency	Avg Rate to INR
Apr-2026	USD	Average USD to INR rate for Apr-2026
Apr-2026	GBP	Average GBP to INR rate for Apr-2026
Apr-2026	INR	1

This ensured that every monthly Budget and Actual value could be converted using the correct monthly average rate.

2. Creation of FX Key

To link the FX rate table with Actuals and Budget, I created a common key called `FX_Key`.

The `FX_Key` was created using:

```
Currency + MonthStart
```

Example:

```
USD|2026-04-01
```

This key was created in all three relevant tables:

- `FX_Rates_Monthly_Avg`
- `Fact_Budget`
- `Fact_Actuals`

The logic was:

```
FX_Key = Currency & "|" & MonthStart
```

This allowed Power BI to correctly match each Actual and Budget row with the right currency rate for that month.

3. Data Modelling

After creating the `FX_Key`, I linked the monthly FX table with the Actual and Budget fact tables.

The relationships used were:

`FX_Rates_Monthly_Avg[FX_Key]` → `Fact_Budget[FX_Key]`

`FX_Rates_Monthly_Avg[FX_Key]` → `Fact_Actuals[FX_Key]`

The relationship was set as:

- Many-To-One
- Single direction
- Active relationship

This means one monthly FX rate is applied to multiple Actual or Budget rows belonging to the same currency and month.

4. Budget Conversion into INR

Once the FX relationship was created, I created a `Budget INR` measure.

```
Budget INR =
SUMX (
    Fact_Budget,
    Fact_Budget[BudgetAmount_num]
        * RELATED ( FX_Rates_Monthly_Avg[Avg_RateToINR_Accounting] )
)
```

This measure works row by row. It takes each Budget amount, finds the matching monthly FX rate using the relationship, converts it into INR, and then sums all converted values.

For example:

- INR budget is multiplied by 1
- USD budget is multiplied by USD to INR rate
- GBP budget is multiplied by GBP to INR rate

This gives one consolidated Budget value in INR.

5. Actual Conversion into INR

Similarly, I converted Actuals into INR using the same FX logic.

```
Actual INR =
SUMX (
    Fact_Actuals,
    Fact_Actuals[Amount_num]
        * RELATED ( FX_Rates_Monthly_Avg[Avg_RateToINR_Accounting] )
)
```

This ensures that Actuals are also converted into INR using the same monthly average FX rate logic.

This step was important because Actual and Budget should be compared in the same reporting currency. Comparing Actual in local currency with Budget in INR would give an incorrect variance.

6. INR-Based Variance Measures

After converting both Actual and Budget into INR, I created INR-based variance measures.

```
Variance INR =
[Actual INR] - [Budget INR]
Variance % INR =
DIVIDE ( [Variance INR], [Budget INR] )
```

This gives a like-for-like comparison because both sides are now in INR.

For this OPEX dashboard:

- Positive variance means Actual INR is higher than Budget INR, which indicates overspend.
- Negative variance means Actual INR is lower than Budget INR, which indicates saving.

7. Dashboard Update

After creating INR-based measures, I updated the Executive Summary dashboard to use the converted INR measures.

The KPI cards were updated to:

- `Actual INR`
- `Budget INR`
- `Variance INR`
- `Variance % INR`

I also updated the monthly trend and variance visuals to use INR-based measures instead of the original local currency measures.

This made the dashboard more accurate because it now compares all values in one reporting currency.

8. Final Result

The final model now converts INR, USD, and GBP values into INR before aggregation. This avoids incorrect summing of mixed currencies and gives a proper consolidated view of Actual, Budget, Variance, and Variance %.

The key logic used was:

- Average FX rate at currency-month level
- Common `FX_Key` using `Currency + MonthStart`
- Relationship between **FX table and fact tables**
- INR conversion using row-level multiplication through `SUMX`
- Final KPI and variance measures based on INR values